

Question:

Which type of articulators permit horizontal as well as vertical motion, but do not orient the motion to the TMJ via facebow transfer?

Answer Choices:

- A. Class I
- B. Class II
- C. Class III
- D. Class IV

Reasoning + Answer:

<tool_call>

To determine which type of articulator allows horizontal as well as vertical motion but does not orient the motion to the TMJ via facebow transfer, we need to understand the classification of dental articulators.

Class I articulators:

- Allow both vertical and horizontal movements.
- Motion is oriented based on facebow transfer, which aligns the articulator with the TMJ.
- Therefore, they do orient motion via facebow transfer.

Class II articulators:

- Allow horizontal and vertical motion.
- The motion is not oriented via facebow transfer; instead, they are designed with a fixed, non-physiologic orientation.
- The jaw movement is not directly tied to the TMJ geometry from the facebow.

Class III articulators:

- Allow only vertical motion.
- No horizontal motion.

Class IV articulators:

- Allow both vertical and horizontal motion.
- However, they are designed to be facebow-oriented, meaning motion is aligned with the TMJ via facebow transfer.

Thus, the only articulator that permits both horizontal and vertical motion but does NOT orient the motion via facebow transfer is Class II.

Class II articulators are known for their ability to simulate horizontal movement (e.g., lateral excursions) without relying on facebow transfer to align with the TMJ. This makes them useful in situations where the TMJ orientation is not critical or when the patient is being evaluated without full TMJ alignment.

Therefore, the correct answer is Class II.

<answer>

B. Class II

</answer>